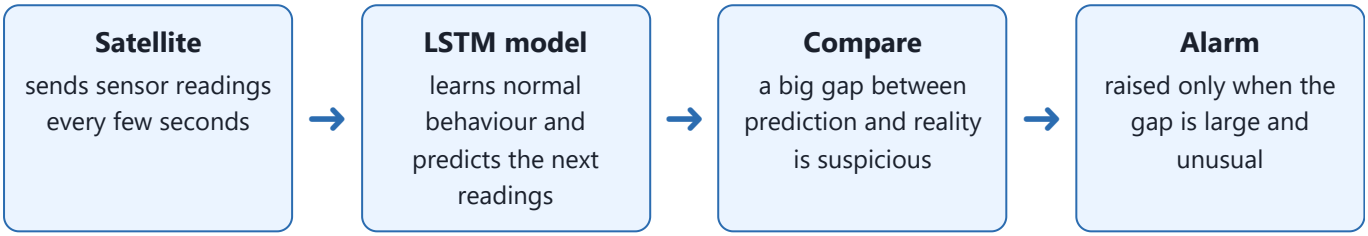
Satellite Health Monitoring with Machine Learning

Team guide · what the project is, what is done, where we stand · November 2026

1. The topic in one minute

A satellite sends thousands of sensor readings (**telemetry**) to the ground, such as temperatures, voltages and currents. Engineers watch for **anomalies**: readings that behave in a way they shouldn't. Missing one can lose a mission, but false alarms waste engineers' time.

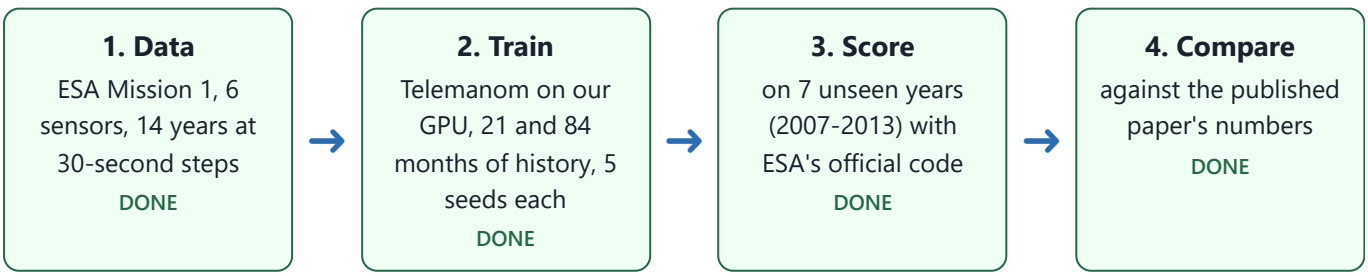
Our project trains a model to spot these anomalies automatically. It then adds smarter decision-making on top: *when* to run the expensive model, *whether* a sensor itself is faulty, *which* alarms belong to the same incident, and *what* to do about it.



Key words

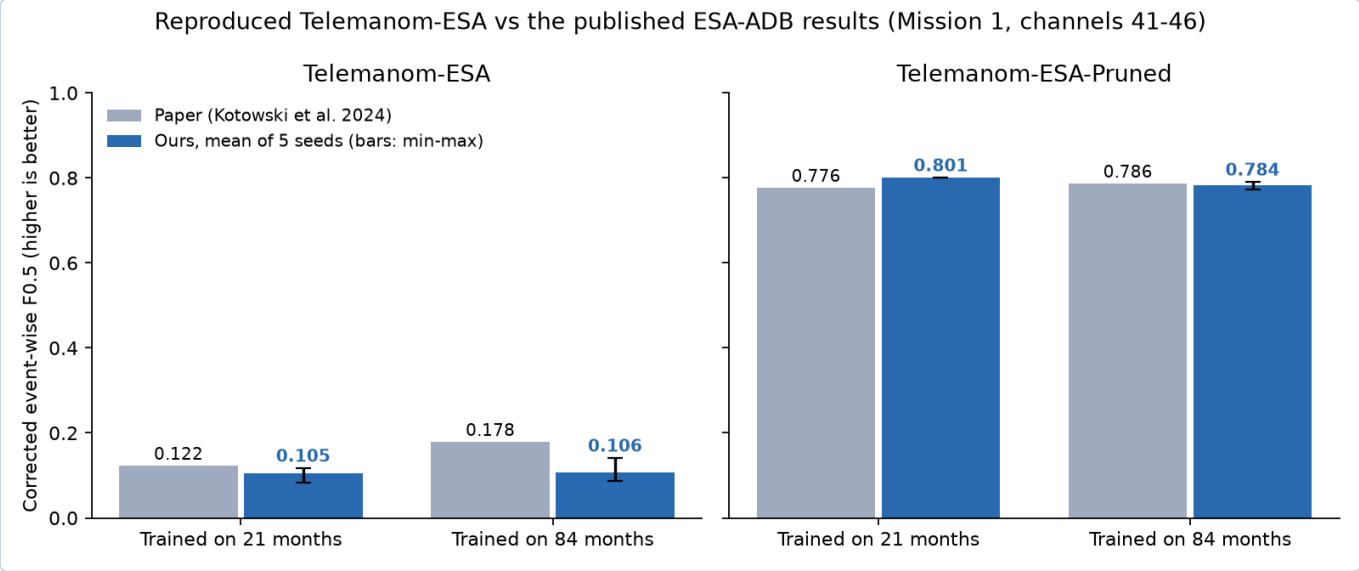
Term	Meaning
ESA-ADB	The European Space Agency's anomaly benchmark: real data from ESA missions, labelled by ESA engineers, with official scoring rules. Our dataset.
Telemanom	The model we trained. An LSTM forecasts each sensor, and an automatic threshold decides when the forecast error counts as an anomaly.
Anomaly / rare event	Anomaly: something is actually wrong. Rare event: unusual but harmless behaviour. The benchmark counts both.
F0.5 score	The main score, from 0 to 1. It treats false alarms as worse than misses, which matches what engineers want.
Pruned	The paper's best variant. It ignores very small forecast errors (below 0.007), which removes most false alarms.
Seed	A random starting point for training. We trained 5 seeds per setting to show the result isn't luck.

2. What has been done



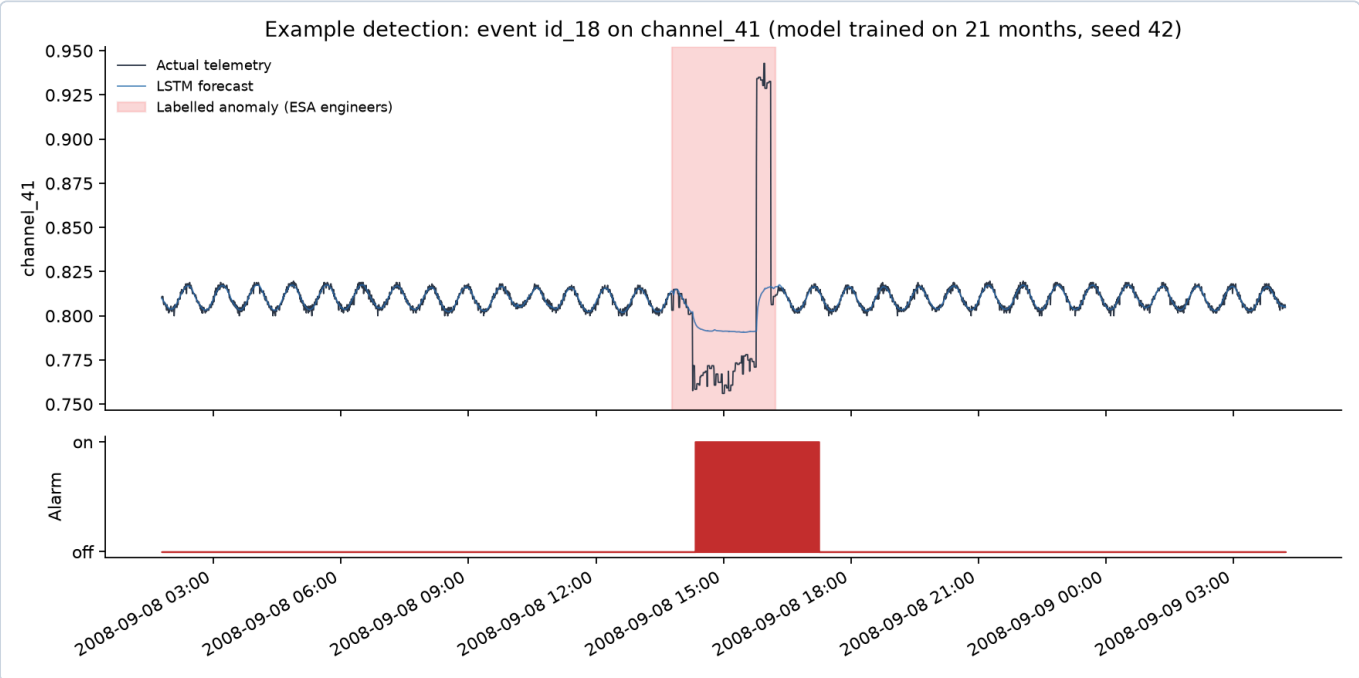
We checked every step against the official numbers: the exact row counts, the 65 test events, and the scoring code against 10 hand-calculated cases.

3. Results: we match or beat the paper

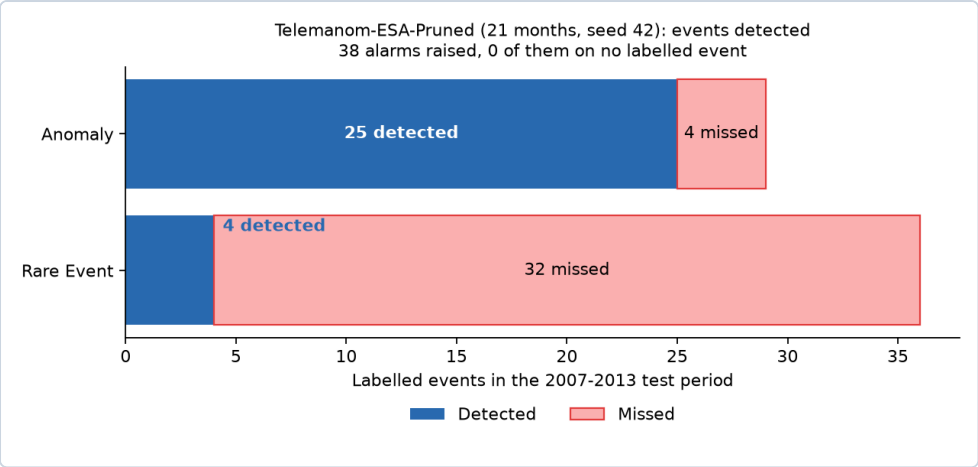


Grey: published paper. Blue: our trained models (average of 5 runs). Higher is better.

Model	Training data	Paper	Ours	Verdict
Telemanom-ESA-Pruned (main)	21 months	0.776	**0.801**	Better (+0.025)
Telemanom-ESA-Pruned (main)	84 months	0.786	**0.784**	Same (−0.002)
Telemanom-ESA (weaker variant)	21 months	0.122	0.105	Slightly lower
Telemanom-ESA (weaker variant)	84 months	0.178	0.106	Lower: more false alarms



Blue: the model's forecast. Red band: a real anomaly labelled by ESA engineers. The forecast stops following the signal, so the alarm (bottom) switches on.



25 of 29 real anomalies caught, and no alarm landed outside a labelled event. Rare events are mostly missed: that is the weakness to fix.

4. Where we stand

P0	P1	P2	P3	P4	P5	P6
Setup and metric check DONE	Reproduce baseline DONE	Fix evaluation rules NEXT	C1 adaptive computation PLANNED	C2 + C3 reliability, graphs PLANNED	C4 response planning PLANNED	C5 ablation, final test PLANNED

We have a **verified baseline**, the "before" that every improvement is measured against.

5. What we will add (our novelty)

Contribution	In simple words	What it should improve
C1 Adaptive computation	A cheap detector checks everything. The heavy LSTM runs only when the cheap one is unsure, the pattern is unfamiliar, or the possible impact is high.	Less compute for the same or better accuracy
C2 Sensor reliability	Tell "the sensor is broken" apart from "the satellite has a real problem", by comparing sensors in the same group.	Fewer false alarms from bad sensors
C3 Incident correlation	Learn which sensors move together, and merge alarms from one incident into a single alert.	Fewer, clearer alerts
C4 Cost-aware response	For each alert, compare options (ignore, watch, escalate) by their expected cost, replayed on the real timeline.	Better decisions, not just detections
C5 Ablation study	Switch each part on and off to prove which one actually helps.	Scientific evidence for the paper

> **Our main finding so far:** the five Pruned models made different predictions but caught exactly the same events. The *decision step* (the threshold) matters more than the neural network. That is exactly where C1-C4 operate, so it supports our direction. The clearest weakness to fix: only 4 of 36 rare events are caught.

6. Related IEEE papers

Paper	Link to our project
Zeng et al., "Satellite Telemetry Data Anomaly Detection Using Causal Network and Feature-Attention-Based LSTM", *IEEE Trans. Instrumentation & Measurement*, 2022	Same idea as our model (LSTM plus a dynamic threshold). Its causal network between sensors inspires C3.
Xu, Cheng, Guo, "A Multivariate Anomaly Detector for Satellite Telemetry Data Using Temporal Attention-Based LSTM Autoencoder", *IEEE Trans. Instrumentation & Measurement*, 2023	Adaptive thresholds and finding which sensor caused an alarm. These link to our decision layer and C2.
Zhang, Hu, "A Fourier Graph Attention Network for Satellite Anomaly Detection", *IEEE CCECE*, 2025	A graph model across sensors, the direction of C3.

7. What to say when presenting

- **Problem:** satellites produce huge amounts of sensor data, and engineers need reliable automatic anomaly alerts with few false alarms.
- **Dataset:** ESA-ADB, real ESA mission data labelled by ESA engineers. It is the official benchmark, so our numbers are comparable to the paper's.
- **Model:** Telemanom, an LSTM forecaster with automatic thresholding, trained by us on our GPU.
- **Result:** F0.5 of 0.801 against the paper's 0.776 (21 months), and 0.784 against 0.786 (84 months). That is the same or better, and stable across 5 training runs.
- **Next:** adaptive computation, sensor reliability, incident correlation and cost-aware response, each tested against this baseline.

Be ready for these questions:

- "*Why is the weaker variant lower?*" It raises more false alarms than the paper's single run did. It catches the same share of events (recall 0.898 against the paper's 0.894). We are investigating, and our main model matches the paper.
- "*Is this your novelty?*" Not yet. This is the reproduced baseline the teacher asked for. The novelty (C1-C4) is built on top of it next.
- *The 6 IEEE papers collected earlier* are about space-debris images and underwater sonar, not satellite telemetry. Present the three papers in Section 6 instead.

8. Where everything is

GitHub: `VONVOLT6/satmon-esa` (private)	What it is
`RESULTS.md`	**The results report to show:** setup, tables, charts, papers
`results/figures/`	All charts as images
`models/`	The 10 trained models (`last.pt` in each folder)
`src/`	The code: data, training, scoring, charts
`README.md`	How to rerun everything